DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CARE OF NEWBORN WITH NEONATAL JAUNDICE
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

CODE AND VERSION NUMBER	
DPP-NICU-0016 (v4)	
LAST VERSION APPROVAL DATE:	28 / 05 / 2023
NUMBER OF PAGES:	7

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APPROVED DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
09 / 02 / 2026	23 / 02 / 2026	TRIENNIAL	08 / 02 / 2029

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1. PURPOSE

1.1. To provide guidelines to health care providers on how to manage sick newborn with neonatal jaundice.

1.2. To emphasize the nursing care to be given to newborn with neonatal jaundice.

2. DEFINITION & ABBREVIATION

2.1. **Jaundice** - is having an SBR (Serum Bilirubin) of above 50 µmo/L per Liter and is the yellowish appearance of the skin that occurs with the deposition of bilirubin in the dermal and subcutaneous tissues and sclera.

2.2. **Bilirubin** – is a yellow substance in the blood that the body creates when it replaces old red blood cells.

2.3. **Hyperbilirubinemia** – the excess of bilirubin in the blood.

2.4. **RH (Rhesus incompatibility)** – occurs during pregnancy if a woman has a RH negative blood and her baby has an RH positive blood that can cause hemolysis.

2.5. **ABO incompatibility** – occurs when mother's blood cells develops antibodies that can attack the newborn's blood cells that cause jaundice and occurs when mother is type O and baby is A, B or AB.

2.6. **Hemolysis** – the destruction of the red blood cells.

2.7. **Phototherapy** – is a treatment for jaundice in the newborn that involves the exposure of the skin to blue light that accelerates the excretion of bilirubin in the skin.

2.8. **TSB (Total Serum Bilirubin Levels)** – measure used when charting serum bilirubin results unto phototherapy charts.

2.9. **SBR (Serum Bilirubin)** – reports the unconjugated and conjugated bilirubin levels.

2.10. **Kernicterus** – the pathogenic diagnosis characterized by bilirubin staining of the brain stem and cerebellum leading to encephalopathy.

2.11. **Bilirubin Encephalopathy** – the acute manifestations of bilirubin toxicity seen in the first few weeks after birth.

2.12. **G6PD Deficiency** – is a condition in which red blood cells are deficient in G6PD enzyme and breaks down when body is exposed to certain drugs, food, stress or infection.

2.13. **WBC** – White Blood Cell

2.14. **OPD** – Outpatient Department

2.15. **ER** – Emergency Room

2.16. **NICU** – Neonatal Intensive Care Unit

3. RESPONSIBILITY

3.1. **Physicians (Consultant, Specialists and Residents)**

3.1.1. Responsible to assess patients for the presence of neonatal jaundice.

3.1.2. Provide a written order of the procedure to manage/treat neonatal jaundice.

3.2. **NICU Nurse (Head Nurse, Charge Nurse and Staff Nurse)**

3.2.1. Responsible in giving nursing care to patient according to physician's ordered treatment for jaundice.

3.2.2. Document any procedures done to patient in the medical file.

4. POLICY

4.1. All newborn patients must be carefully assessed and evaluated for neonatal jaundice in Pediatric Emergency Room, Neonatology OPD and Nursery and will be subjected for admission to Pediatric ward and Neonatal Intensive Care Unit based on the newborn's clinical condition and physician's written order.

4.2. Assessment criteria of patient for Neonatal Jaundice

4.2.1. Patients must be carefully identified who can develop significant hyperbilirubinemia that have the following risk factors

4.2.1.1. Gestational age under 38 weeks or prematurity

4.2.1.2. A previous sibling with neonatal jaundice requiring phototherapy

4.2.1.3. A family history of G6PD deficiency

4.2.1.4. Mother's intention to breastfeed exclusively

4.2.1.5. Mother with positive antibody screen

4.2.1.6. Cephalohematoma, bruising and trauma from instrumental birth

4.2.1.7. Visible jaundice in the first 24 hours of life.

4.2.1.8. Dehydration

4.2.1.9. Delayed passage of meconium

4.2.1.10. ABO or RH incompatibility

4.2.1.11. Inadequate breastfeeding

4.3. The evaluation of jaundiced newborn must include a thorough history and physical examination with emphasis on the state of hydration and consideration of the possibility of an acute hemolytic process or infection.

4.4. Visually jaundiced well babies must be checked through transcutaneous bilirubinometer (jaundice meter) to determine bilirubin levels in newborn equal or above 38 weeks and newborn with gestational age in between 34-37 weeks, who are above 72 hours old to

determine the need for a laboratory serum bilirubin.

4.5. Initial TSB/SBR measurement will be requested based on clinical observation and the following factors:

4.5.1. Any neonate <24 hours' age with clinically apparent jaundice

4.5.2. Any neonate where there is clinical doubt about the degree if jaundice

4.5.3. Any neonate with risk factors for jaundice (ABO/RH incompatibility, sepsis, acidosis, asphyxia, hypoalbuminemia)

4.6. On-going TSB/SBR measures should be requested or repeated at intervals depending on the initial level and rate of rise.

4.7. A TSB/SBR may be requested within 24 hours of ceasing phototherapy to assess for rebound hyperbilirubinemia and neonate at increased risk of clinically significant rebound hyperbilirubinemia including those born less than 37 weeks' gestation, those not feeding optimally or those with hemolytic disease.

4.7.1. Further blood investigations includes:

4.7.1.1. Maternal and infant blood type

4.7.1.2. Direct coombs test

4.7.1.3. Full blood count for hemoglobin, hematocrit, platelet count and WBC differential.

4.7.1.4. Urinalysis for reducing substances

4.7.1.5. Septic Screen if sepsis suspected

4.7.1.6. G6PD and galactosemia screens if suspected

4.7.1.7. Serum thyroxin and Thyroid-stimulating hormone levels

4.7.2. Ensure that newborn screening test was done

4.8. Depending on the patient's bilirubin level and by other underlying condition, treatment may consist of both ensuring adequate hydration and encouraging photo-degradation of unconjugated bilirubin in the skin by the provision of phototherapy that can prevent serum bilirubin levels to rise to levels where exchange transfusion would be necessary.

4.9. Commence phototherapy once TSB/SBR is greater than the appropriate reference range for neonate's gestation/weight and presence of risk factors.

4.10. **Use of Phototherapy**

4.10.1. **Single Phototherapy for Term Babies**

4.10.1.1. Use conventional blue light phototherapy as treatment for significant hyperbilirubinemia in babies with gestational age of 37 weeks or more unless

4.10.1.2. The serum bilirubin level is rising rapidly (more than 8.5 micromole/L per hour)

4.10.1.3. The serum bilirubin is at a level that is within 50 micromoles/L below the threshold for which exchange transfusion is indicated after 72 hours.

4.10.2. **Single Phototherapy for Preterm Babies**

4.10.2.1. Use either fiberoptic phototherapy or conventional "blue light" phototherapy as treatment for significant hyperbilirubinemia in babies less than 37 weeks unless

4.10.2.2. The serum bilirubin level is rising rapidly (more than 8.5 micromole/L per hour).

4.10.2.3. The serum bilirubin is at a level that is within 50 micromoles/L below the threshold for which exchange transfusion is indicated after 72 hours.

4.10.3. **Continuous Multiple Phototherapy for Term/Preterm Babies**

4.10.3.1. Initiate continuous multiple phototherapy to treat all babies if any of the following apply:

4.10.3.1.1. The serum bilirubin level is rising rapidly (more than 8.5 micromole/L per hour).

4.10.3.1.2. The serum bilirubin is at level within 50 micromoles/L below the threshold for which exchange transfusion is indicated after 72 hours.

4.10.3.1.3. The bilirubin level fails to respond to single phototherapy (that is the level of serum bilirubin continues to rise, or does not fall, within 6 hours of starting single phototherapy.

4.10.4. **Capsule phototherapy** – this therapy should be initiated when the total serum bilirubin reaches the critical level of exchange transfusion.

4.10.5. **Stopping phototherapy**

4.10.5.1. Stop phototherapy once serum bilirubin has fallen to a level at least 50 micromoles/L below the phototherapy threshold.

4.10.5.2. Check for rebound of significant hyperbilirubinemia with a repeat serum bilirubin measurement 12-18 hours after stopping phototherapy.

4.10.6. **TSB Monitoring while on Phototherapy**

4.10.6.1. Patients on single phototherapy must be monitored for TSB level once daily or as directed by physician

4.10.6.2. Patients on Double Phototherapy must be monitored for TSB level twice

daily or as directed by physician

4.10.6.3. Patients on capsule phototherapy must be monitored for TSB level after 4 hours of exposure and as directed by physician.

4.11. **Intravenous Immunoglobulin**

4.11.1. Use Intravenous Immunoglobulin (IVIG) 500mg/kg over 6 hours) as an adjunct to continuous multiple phototherapy in cases of Rhesus hemolytic disease or ABO hemolytic disease when serum bilirubin continues to rise by more than 8.5 micromole per liter per hour.

4.12. **Fluids and Nutrition**

4.12.1. The total fluid intake (TFI) for a 24-hour period may need to be increased by at least 10% (10-15 mL/kg/day for each unit of phototherapy) to account for insensible fluid loss when a neonate is receiving phototherapy.

4.12.2. Parenteral nutrition and intravenous fluids should continue as ordered and may also need to be increased by 10% to account for insensible fluid loss.

4.12.3. Breastfeed babies who require phototherapy should continue to breastfeed unless contraindicated due to other pathology.

4.12.4. The neonate's sucking: attachment and mother's milk supply should be monitored.

4.12.4.1. Neonates who are receiving enteral feeds or expressed breast milk or infant formula should continue to do so.

4.13. An exchange transfusion will be indicated for any infant in which the degree of hyperbilirubinemia is severe that cannot be adequately controlled by phototherapy alone.

4.14. Patients placed under phototherapy must be carefully monitored for:

4.14.1. Adequacy of hydration (urine output)

4.14.2. Adequacy of nutrition (weight gain)

4.14.3. Temperature (hypo/hyperthermia)

4.14.4. Skin rashes

4.14.5. Loose stools

4.14.6. Potential signs of bilirubin encephalopathy

4.15. For the guidelines on phototherapy refer to DPP-NICU-0007 Care of Patient on Phototherapy.

5. PROCEDURE

5.1. The physician will assess the patient for neonatal jaundice.

5.2. The physician will write the order of the procedure in the medical file according to this management.

5.3. The nurse will verify and carry out the physician's order.

5.4. Perform hand hygiene and wear appropriate PPE according to the procedure to be done.

5.5. Identify the patient correctly.

5.6. The nurse will perform the procedure to manage neonatal jaundice (i.e. phototherapy, IV infusion, antibiotic therapy) according to policy and as per physician's order.

5.7. The nurse will give the appropriate nursing care for jaundice patient according to physician's order and management.

5.8. Document the procedure done to patient in the medical file.

6. MEASUREMENT

6.1. 100% of the staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0016(v3): Care of Newborn with Neonatal Jaundice, 28/05/2023, p. 1-7

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10 p.126-127

7.3. MOH-KSA Guidelines for Neonatal Care

8. ATTACHMENT

8.1. Bilirubin Thresholds Graphs

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.